Recognizing Human Activities in Video

Using Local Features and Naive Bayes

Recognizing image ‘‘style’’ and activities in video
using local features and naive Bayes
Daniel Keren - Elsevier 2003

Avi Guttman

The Original Model

Paper-faithful implementation - DCT + Naive Bayes

How it works

- Videos preprocessed to 64×64 grayscale frames
- Slide a 5×5×5 window (length × width × depth-time)
- Motion filter - skip static background blocks
- 3D DCT → 125-dim feature vector per block
- Top 10 features by Mutual Information
- Naive Bayes: binary thresholds + log-probabilities
- Confidence filter: winner must score ≥ 2× loser

Pipeline

MP4 videos

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64×64 grayscale frames

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5×5×5 block extraction

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3D DCT → 125-dim vector

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MI feature selection (top 10)

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Naive Bayes classification

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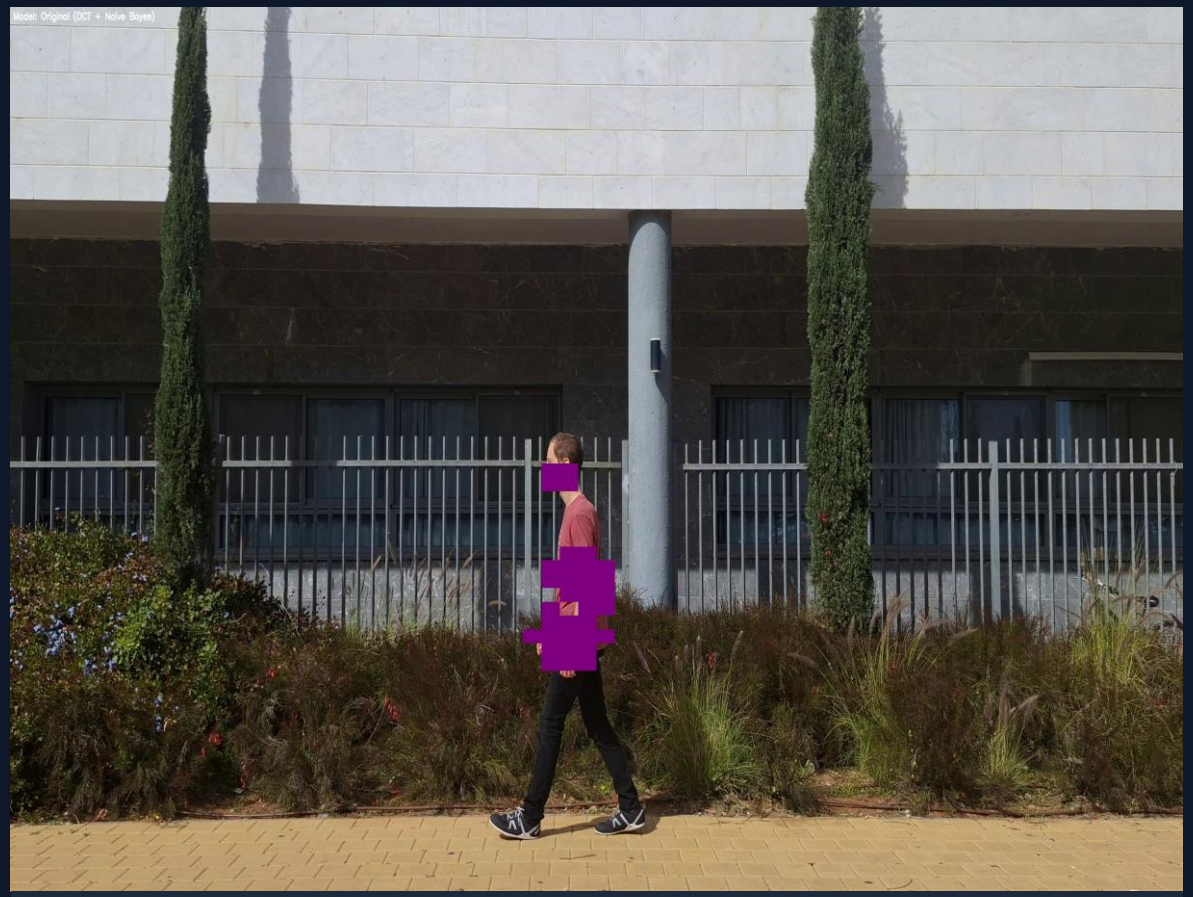
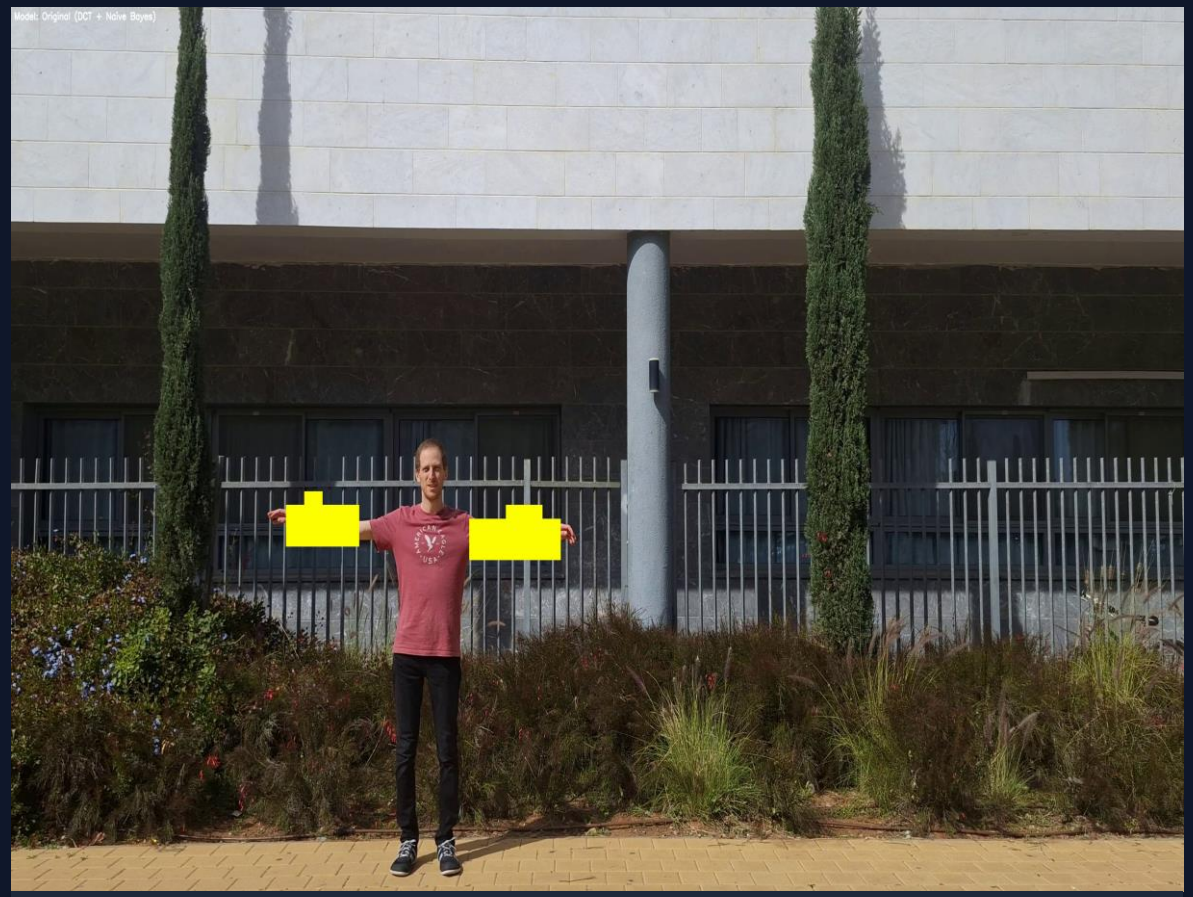
Labeled output video

Original Model - Results

Per-pixel activity localization on test video



Walking - purple spread on all body



Waving - yellow on hands

Body-part level localization · Confidence filter removes low-certainty detections

The Improved Model

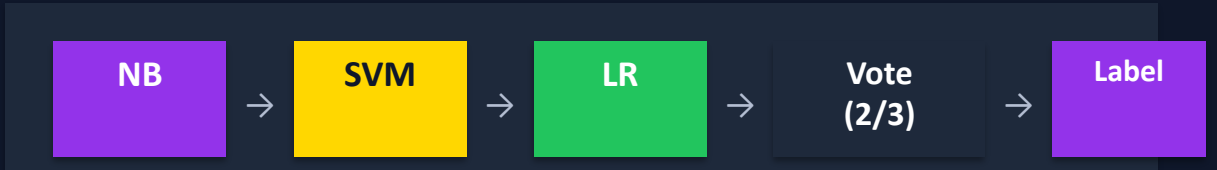
Ensemble of 3 classifiers at 128×128 resolution

Key changes

- 128×128 resolution (was 64×64)
  - 5× more training data per frame
- Data augmentation - horizontal flip
  - Doubles the training set
  - Robustness
- 3 classifiers instead of 1:
  - Naive Bayes (same approach, larger resolution)
  - SVM with RBF kernel (C=1)
  - Logistic Regression (L2, C=1)
- Majority vote: pixel labeled if 2 of 3 agree

Post-processing

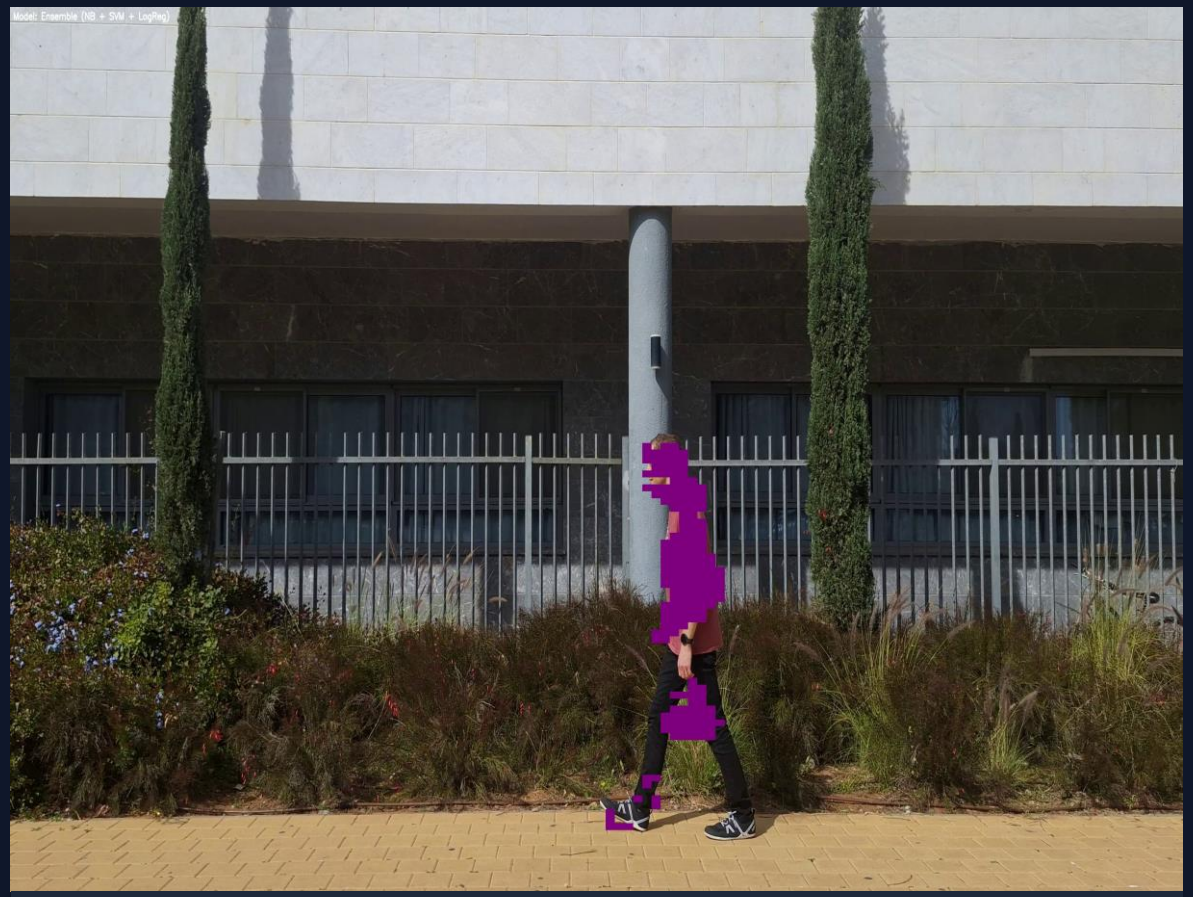
- Temporal smoothing - 5-frame majority window
  - Removes per-frame label flicker
- Dominant-class suppression (45% threshold)
  - If < 45% of pixels are waving → erase all waving
  - If > 55% of pixels are waving → erase all walking
- Blob cleanup - remove connected components < 60px



Majority vote ensemble

Improved Model — Results

Ensemble (NB + SVM + LogReg) on test video



Walking - large purple blob on person

Waving - large yellow blobs on arms

Person-level detection · Temporal smoothing + Blob cleanup reduce noise

Improvements & Findings

What I changed	Effect
64×64 → 128×128 resolution	Biggest win - 5× more training samples per frame
Data augmentation (horizontal flip)	Doubles dataset, adds view robustness
SVM (RBF, C=1) added to ensemble	Better then Naive Bayes alone
Logistic Regression added to ensemble	Added classifier diversity to the vote
Temporal smoothing (5-frame window)	Eliminates frame-to-frame label flicker
Dominant-class suppression (45%)	Cleaner per-frame labels, less cross-contamination

Key findings

- Resolution was the most impactful single change - more pixels per block = better texture descriptors
- Ensemble did not dramatically outperform a single model (features are correlated → errors are correlated)
- Arm swing during walking still misclassified as waving - probably fundamental DCT feature limitation

Summary & Conclusions

- Implemented Keren (2003) faithfully - DCT + Naive Bayes for per-pixel activity localization
- Extended with an ensemble model (NB + SVM + LogReg) at 128×128 resolution
- Higher resolution was the most effective improvement
- Post-processing (smoothing + dominant suppression) significantly improved visual quality
- Classical signal processing methods can still produce meaningful results without deep learning
- Main limitation: DCT features cannot distinguish arm-swing from waving - a fundamental constraint

Thank you